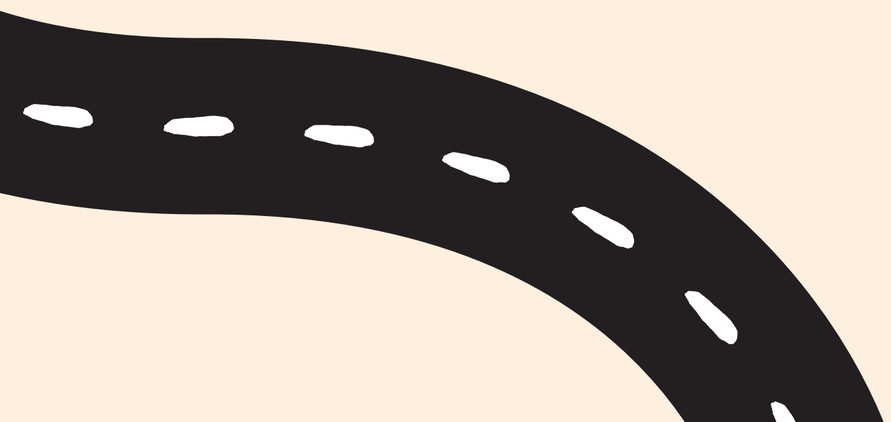
PUBLIC TRANSPORT SYSTEM OF MILAN: A KNOWLEDGE GRAPH APPROACH

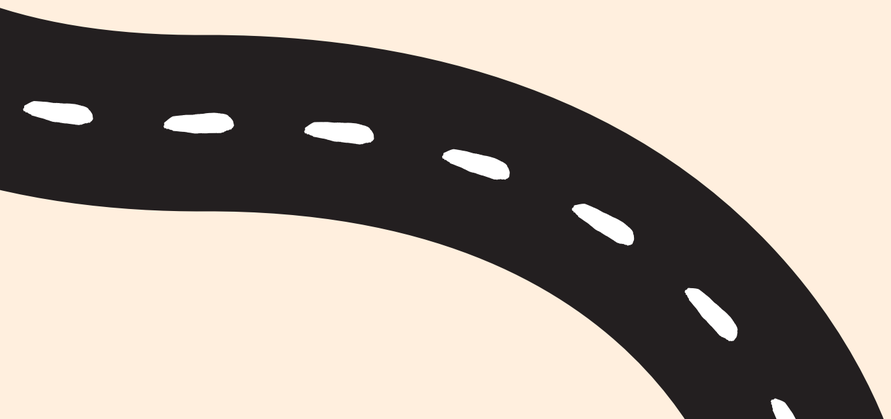
Transforming GTFS Data for Complex Semantic Queries

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THE CHALLENGE



LIMITS OF STANDARD DATA

Urban transport systems are **complex networks** (stops, routes, schedules, fare zones).

GTFS provides a **standard** format, but acts like a relational database.

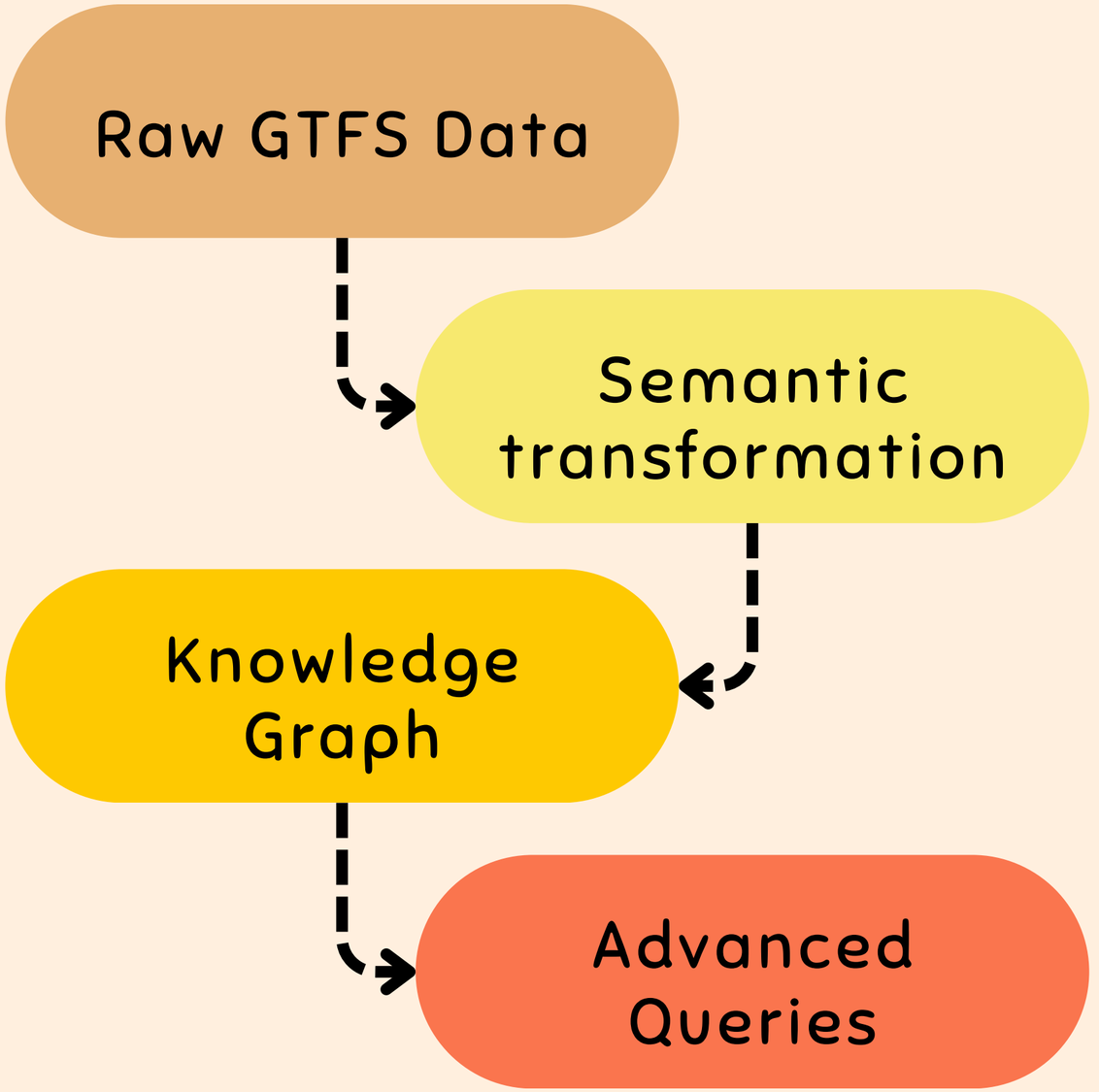
Limitation: Lacks explicit, contextual relationships.

OBJECTIVE

01 **Transform** raw GTFS data into a dynamic Knowledge Graph.

02 **Enrich** data with missing contextual information.

03 Enable intelligent, context-aware **queries**.



DATA SOURCE AND SPECIFICATION

GENERAL TRANSIT FEED SPECIFICATION (GTFS)

Comune di Milano

- **agency**: transit agency (ATM).
- **calendar**: regular service patterns.
- **calendar_dates**: exceptions to the service.
- **feed_info**: metadata about the dataset.
- **routes**: transit routes.
- **shapes**: physical paths vehicles take
- **stop_times**: arr/dep times for each stop on each trip.
- **stops**: list of all stops
- **transfers**: rules for transferring.
- **trips**: trips and schedule details.

DATA PREPROCESSING AND ENRICHMENT

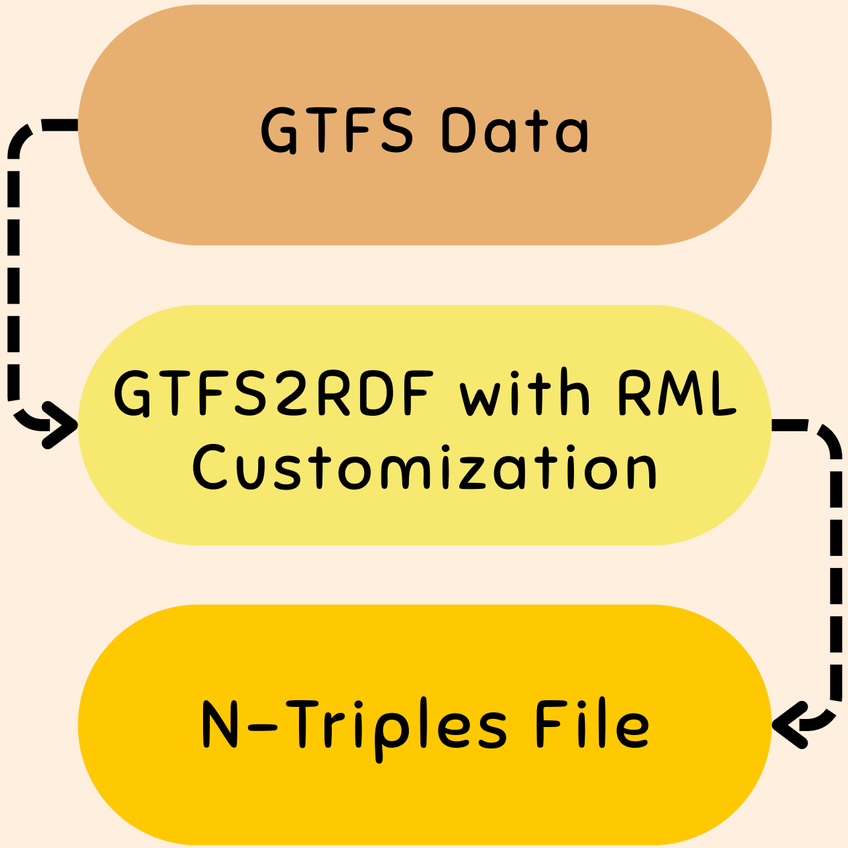
01 PERFORMANCE OPTIMAZATION

Strategically **sampled** the massive **stop_times.txt** file for efficient processing.

02 DATA ENRICHMENT

Problem: original data was missing fare zones.
Solution: integrated external GeoJSON from Milan Open Data for zone boundaries. Developed a Python script to perform point-in-polygon analysis.
Result: every stop was tagged with its correct zone.

SEMANTIC MAPPING



A TOOL USED

GTFS2RDF: Open-source converter.
RML File: gtfsde-rml.ttl defined rules for conversion.

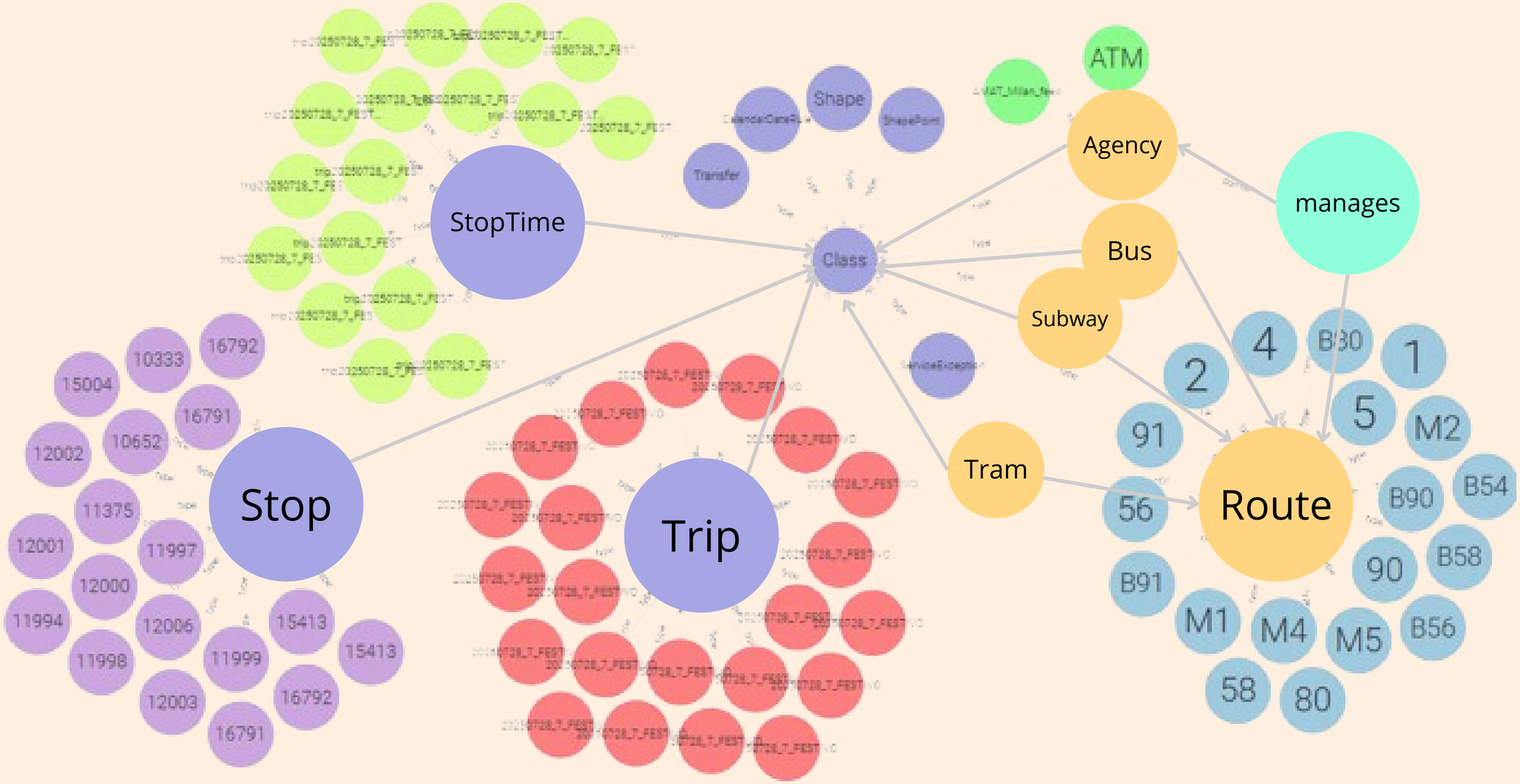
B CUSTOMIZATIONS

Localization: from German (de) to Italian (it)
Data Enrichment: Ensured each stop is linked to its correct zone.

C OUTPUT

A single N-Triples **(.nt) file** containing the complete graph.

HOSTING AND VISUALIZATION



QUERY 1: DIRECT TRIPS BETWEEN TWO STOPS

Goal: find all direct trips between stop A and B.
Function: Core pathfinding and scheduling.
Result: list of all direct trips with departure times.

First Query | Second Query | Third Query

```
PREFIX gtfs: <http://vocab.gtfs.org/terms#>
PREFIX foaf: <http://xmlns.com/foaf/0.1/>

SELECT DISTINCT ?trip ?departureTime
WHERE {
  # StopTime per la fermata di partenza
  ?st1 a gtfs:StopTime ;
       gtfs:trip ?trip ;
       foaf:stop <http://data.gtfs.org/gtfs.de/2021-03-08/stops/15270> ;
       gtfs:departureTime ?departureTime .

  # StopTime per la fermata di arrivo (stesso trip)
  ?st2 a gtfs:StopTime ;
       gtfs:trip ?trip ;
       foaf:stop <http://data.gtfs.org/gtfs.de/2021-03-08/stops/11136> .
}
ORDER BY ?departureTime

```

QUERY 2: ALL ROUTES IN A SPECIFIC ZONE

Goal: list all routes serving an area.
Function: spatial aggregation and urban planning analysis.
Key Advantage: this was impossible with the original dataset.

First Query | Second Query | Third Query

```
PREFIX gtfs: <http://vocab.gtfs.org/terms#>
PREFIX foaf: <http://xmlns.com/foaf/0.1/>

SELECT ?route (SAMPLE(?shortName) AS ?uniqueShortName) (SAMPLE(?longName) AS ?uniqueLongName
WHERE {
  ?stop a gtfs:Stop ;
        gtfs:zone ?zoneId .
  FILTER(LCASE(STR(?zoneId)) = "baggio")

  ?stopTime a gtfs:StopTime ;
            foaf:stop ?stop ;
            gtfs:trip ?trip .

  ?trip gtfs:route ?route .

  OPTIONAL { ?route gtfs:shortName ?shortName }
  OPTIONAL { ?route gtfs:longName ?longName }
}
GROUP BY ?route
ORDER BY ?uniqueShortName
```

QUERY 3: DIRECT ALTERNATIVE ROUTES

Goal: enumerate all unique routes connecting 2 points
Function: service discovery and route suggestion for users.
Result: list of all available direct lines.

First Query | Second Query | Third Query

```
PREFIX gtfs: <http://vocab.gtfs.org/terms#>
PREFIX foaf: <http://xmlns.com/foaf/0.1/>

SELECT ?route (SAMPLE(?shortName) AS ?ShortName) (SAMPLE(?longName) AS ?LongName)
WHERE {

  ?stopTimeA a gtfs:StopTime ;
             foaf:stop <http://data.gtfs.org/gtfs.de/2021-03-08/stops/15270> ;
             gtfs:trip ?trip ;
             gtfs:stopSequence ?seqA .

  ?stopTimeB a gtfs:StopTime ;
             foaf:stop <http://data.gtfs.org/gtfs.de/2021-03-08/stops/11136> ;
             gtfs:trip ?trip ;
             gtfs:stopSequence ?seqB .
  FILTER(?seqA < ?seqB)
  ?trip gtfs:route ?route .
  OPTIONAL { ?route gtfs:shortName ?shortName }
  OPTIONAL { ?route gtfs:longName ?longName }
}
GROUP BY ?route
ORDER BY ?ShortName
```

CONCLUTION

A FOUNDATION FOR A SMARTER CITY

01 **Successfully transformed** relational GTFS data into a semantic Knowledge Graph.

02 **Enriched** the model with fare zone data, unlocking new analyses.

03 **Demonstrated** the graph's power to answer complex, real-world questions.

WHAT'S NEXT?

Real-time Data Integration

User-Facing Application

Multi-modal Transit

